

# What is the basic structure of a C++ program?

## Structure of a C++ program

A **C++ program** is structured in a specific and particular manner. In C++, a program is divided into the following three sections:

1. Standard Libraries Section
2. Main Function Section
3. Function Body Section

For example, let's look at the implementation of the **Hello World** program:

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello World!" << endl;
    return 0;
}
```

## 1. Standard libraries section

```
#include <iostream>
using namespace std;
```

- **#include** is a specific preprocessor command that effectively copies and pastes the entire text of the file, specified between the angle brackets, into the source code.
- The file **<iostream>**, which is a standard file that should come with the C++ compiler, is short for **input-output streams**.

This command contains code for displaying and getting an input from the user.

- `namespace` is a prefix that is applied to all the names in a certain set. `iostream` file defines two *names* used in this program - **`cout`** and **`endl`**.
- This code is saying: Use the `cout` and `endl` tools from the `std` toolbox.

## 2.Main function section

```
3. int main() {}
```

4. The starting point of all C++ programs is the `main` function.
5. This function is called by the operating system when your program is executed by the computer.
6. `{` signifies the start of a block of code, and `}` signifies the end.

## 4.Function body section

```
cout << "Hello World" << endl;  
return 0;
```

- The name `cout` is stands for **character output** and displays whatever is between the `<<` brackets.
- Symbols such as `<<` can also behave like functions and are used with the keyword `cout`.
- The `return` keyword tells the program to return a value to the function `int main`
- After the return statement, execution control returns to the operating system component that launched this program.
- Execution of the code terminates here.

## Skeleton of C++ Program

